

MAT11-3

Linear Algebra I

Matrices, vector spaces, and linear maps

One idea — linearity — recognised across many objects and computed with matrices.

Albert School — 38 hours

Preface

In **Foundations** (MAT11-1) you assembled the toolkit this course depends on, and three of its lessons matter most here. First, **how to reason**: you learned to read an “if P then Q ” statement, to prove a universal claim by taking an arbitrary element, and to demolish a false one with a single counterexample. Linear algebra is the first course where you will do this constantly, on your own, and we open by sharpening exactly those reflexes. Second, **functions**: you learned what it means for a map to be injective, surjective, bijective, and how its image $f(A)$ and preimage $f^{-1}(B)$ behave. A linear map is just a function that respects addition and scaling, and every question we ask about it — is it injective? surjective? — is the same question you already met, now with a powerful extra structure that makes it answerable by counting. Third, the concrete objects you computed with — **numerical tuples, polynomials, functions** — will turn out to be instances of one single abstract notion, the **vector space**.

That last point is the soul of the course. You have been adding polynomials, scaling functions, and combining numerical vectors as if these were unrelated skills. They are not. Each of these worlds obeys the **same** short list of rules, and once we name those rules we can prove a theorem once and have it hold in all of them at once. This is the recurring pleasure of linear algebra: a fact you establish about anonymous “vectors” instantly becomes a fact about polynomials, about matrices, about solutions of differential equations you have not even met yet.

The course travels in a deliberate arc, from the concrete to the abstract and back. We begin with the object you can touch and compute with — the **matrix** — and the most ancient problem it solves, a **system of linear equations**. From the mechanical act of solving such a system (**Gaussian elimination**) every later abstraction will crystallize: the pivots will become **dimension**, the free variables will become a **basis** of solutions, and the coefficient matrix will reveal itself as a **linear map**. We then make the leap to the abstract definitions — **vector space, subspace, linear independence, basis, dimension** — and find, reassuringly, that the row reduction we already know how to do is exactly the tool that answers every abstract question. Finally we study **linear maps** in their own right, prove the theorem that governs them all (the **rank theorem**), and close the circle by encoding any linear map back into a matrix.

Keep one slogan in view throughout: **to be linear is to commute with linear combinations**. Everything — matrices, subspaces, bases, the rank theorem — is a consequence of taking that one idea seriously.

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1 Proof literacy

Before any linear algebra, a word on **how we will argue**. This course asks you not just to compute but to **justify** — to prove that a set is a subspace, or to disprove that some vectors are independent. You met the underlying logic in MAT11-1; here we isolate the three moves you will use on almost every page, so that later, when a proof says “we verify the three criteria”, you know exactly what is being asked.

1.1 Direct verification of a universal statement

Most claims in this course have the shape “**for all** x , $P(x)$ ”. To prove such a claim you may not test a few examples and stop — you must take a **completely arbitrary** x , assume only what the hypotheses give you, and derive $P(x)$ using definitions alone.

Worked example — the shape of a direct proof. Claim: for every real number x , $x^2 - 2x + 1 \geq 0$. We take an arbitrary $x \in \mathbb{R}$ — not $x = 3$, not “a few values”, but an unnamed one standing for all of them — and observe $x^2 - 2x + 1 = (x - 1)^2$, a square, hence ≥ 0 . Because x was arbitrary, the conclusion holds for every real number. The **subspace test** in Chapter 7 is verified in precisely this style.

1.2 Disproof by counterexample

To **refute** a universal claim “for all x , $P(x)$ ” you need exactly one x for which $P(x)$ fails. A single well-chosen counterexample is a complete proof of falsity — no more is required, and no amount of supporting examples would help the opposite case.

Worked example — one counterexample is enough. Claim: “for all matrices A, B , $AB = BA$.” We will see in Chapter 3 that this is **false**, and we kill it with one pair of 2×2 matrices for which $AB \neq BA$. We do not need to understand **why** multiplication is non-commutative in general; producing the single failing instance settles it.

Common pitfall The two moves are not interchangeable. Examples can only **disprove** a “for all” statement, never **prove** it. Verifying $AB = BA$ for ten random pairs establishes nothing about all pairs. Conversely, to disprove “there exists x with $P(x)$ ” you must rule out **every** x — a universal argument, not an example. Match the move to the quantifier.

1.3 Reading an “if–then” proof

A statement “**if** P **then** Q ” is proved by **assuming** P and **deriving** Q . Its **contrapositive** “if not Q then not P ” is logically equivalent (MAT11-1), so we may prove either form, whichever is more convenient. We will lean on this when, for instance, proving “if f is injective then $\ker f = \{0\}$ ” is awkward but its contrapositive is immediate.

These three reflexes — verify universally, refute by example, assume-then-derive — are assumed silently in every proof that follows.

2 Matrix arithmetic

We start with the object you can hold in your hand and compute with directly. A **matrix** is a rectangular table of numbers; everything abstract later in the course is, ultimately, a statement about matrices in disguise.

2.1 Definition and notation

A **matrix** of size $m \times n$ is a rectangular array of real numbers with m rows and n columns,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

written compactly $A = (a_{ij})$, where a_{ij} is the entry in **row i , column j** . The set of all such matrices is written $M_{m,n}(\mathbb{R})$, and $M_n(\mathbb{R})$ when $m = n$ (the **square** matrices). A matrix with one column ($n = 1$) is a **column vector**; with one row ($m = 1$), a **row vector**.

Common pitfall The index order is **row first, then column** — always. The entry a_{23} lives in row 2, column 3, never the reverse. A matrix of size $m \times n$ has m rows and n columns, in that order. Getting this backwards is the single most common source of confusion in matrix multiplication below.

2.2 The operations

Before the general rules, compute one concrete instance of each.

Worked example — all four operations on small matrices. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -1 \\ 2 & 5 \end{pmatrix}$. Then

$$A + B = \begin{pmatrix} 5 & 1 \\ 2 & 8 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & 6 \\ 0 & 9 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}.$$

For the product, each entry of AB is a row of A “dotted” against a column of B :

$$AB = \begin{pmatrix} 1 \cdot 4 + 2 \cdot 2 & 1 \cdot (-1) + 2 \cdot 5 \\ 0 \cdot 4 + 3 \cdot 2 & 0 \cdot (-1) + 3 \cdot 5 \end{pmatrix} = \begin{pmatrix} 8 & 9 \\ 6 & 15 \end{pmatrix}.$$

Definition 1 Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices.

- **Addition** (defined only when A and B have the **same size**): $(A + B)_{ij} = a_{ij} + b_{ij}$.
- **Scalar multiplication**: for $\lambda \in \mathbb{R}$, $(\lambda A)_{ij} = \lambda a_{ij}$.
- **Transpose**: A^T is the $n \times m$ matrix obtained by turning rows into columns, $(A^T)_{ij} = a_{ji}$.
- **Matrix multiplication** (defined only when A is $m \times n$ and B is $n \times p$, so the inner sizes match): AB is the $m \times p$ matrix

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

The product entry $(AB)_{ij}$ pairs the i -th **row** of A with the j -th **column** of B , multiplying corresponding entries and summing. For this pairing to make sense the row of A and the column of B must have the same length: **the number of columns of A must equal the number of rows of B** .

Common pitfall Matrix multiplication is the operation students rush. Two traps: (i) the product AB is only defined when the inner dimensions agree — $(m \times n)(n \times p)$ — and its size is the **outer** dimensions $m \times p$; (ii) you sum **across** a row of A and **down** a column of B , not entry by matching entry. When in doubt, write the dimensions under the matrices and cancel the inner pair.

2.3 Algebraic properties — and the one that fails

Proposition 2 For matrices of compatible sizes and scalars $\lambda, \mu \in \mathbb{R}$:

- addition is **commutative** and **associative**, with the zero matrix as identity element;
- scalar multiplication distributes: $\lambda(A + B) = \lambda A + \lambda B$ and $(\lambda + \mu)A = \lambda A + \mu A$;
- multiplication is **associative**, $(AB)C = A(BC)$, and **distributes** over addition, $A(B + C) = AB + AC$;
- the transpose reverses products: $(AB)^T = B^T A^T$.

Every one of these matches your arithmetic instincts — **except** one crucial absence from the list. Matrix multiplication is **not commutative**.

Proposition 3 In general $AB \neq BA$.

Proof. By counterexample, exactly as Chapter 1 prescribes. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so $AB \neq BA$. One pair suffices to refute the universal claim. $\square$

Common pitfall Non-commutativity is not a curiosity — it is a fact you must **actively remember** in every manipulation. You may not “swap factors” as with numbers. In particular $(A + B)^2 = A^2 + AB + BA + B^2$, which is **not** $A^2 + 2AB + B^2$ unless $AB = BA$. And $(AB)^T = B^T A^T$ reverses the order — writing $A^T B^T$ is the usual slip.

2.4 The identity matrix

The $n \times n$ **identity matrix** I_n has 1 on the main diagonal and 0 everywhere else,

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is the neutral element of multiplication: $AI_n = A$ and $I_n A = A$ whenever the product is defined. It plays the role the number 1 plays for ordinary multiplication, and it will reappear as the matrix of the “do-nothing” linear map.

3 Linear systems

The oldest use of matrices is to **solve simultaneous linear equations**. This chapter turns the schoolbook task of solving such a system into a systematic algorithm whose by-products — pivots, free variables — become the backbone of the whole course.

3.1 Matrix form

A system of m linear equations in n unknowns $x_1, \dots, x_n$,

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n = b_m \end{cases},$$

collapses into a single **matrix equation**

$$AX = B,$$

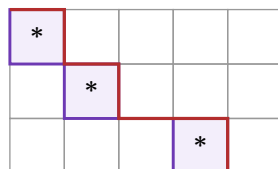
where $A = (a_{ij}) \in M_{m,n}(\mathbb{R})$ is the **coefficient matrix**, $X = (x_1, \dots, x_n)^T$ the column of unknowns, and $B = (b_1, \dots, b_m)^T$ the column of right-hand sides. The product AX , read with the rule of Chapter 3, reproduces the left-hand sides exactly — which is the whole reason matrix multiplication is defined the way it is.

3.2 Gaussian elimination

The method is the one you already use by hand — eliminate a variable, repeat — made systematic. We work on the **augmented matrix** $(A \mid B)$, the coefficient matrix with the right-hand side appended as an extra column, and we are allowed three **elementary row operations**, each of which transforms the system into an **equivalent** one (same solution set):

1. **swap** two rows;
2. **scale** a row by a non-zero constant;
3. **add** a multiple of one row to another.

The goal is **row echelon form**: in every non-zero row the first non-zero entry — the **pivot** — sits strictly to the right of the pivot in the row above, and any all-zero rows are pushed to the bottom. The pivots then trace a descending “staircase”.



pivots descend left-to-right

Figure 1: Row echelon form: each pivot (purple, marked *) lies strictly to the right of the one above it, so the pivots descend in a staircase (red). Columns **with** a pivot correspond to pivot variables; columns **without** one — here the 3rd and 5th — correspond to free variables.

Worked example — elimination to echelon form. Solve

$$\begin{cases} x + 2y + z = 4 \\ 2x + 5y - z = 3 \\ x + 3y - 2z = -1 \end{cases}.$$

The augmented matrix, then $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 4 \\ 2 & 5 & -1 & 3 & 3 \\ 1 & 3 & -2 & -1 & -1 \end{array} \right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & 4 \\ 0 & 1 & -3 & -5 & -5 \\ 0 & 1 & -3 & -5 & -5 \end{array} \right).$$

Now $R_3 \leftarrow R_3 - R_2$ wipes out the third row entirely:

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 4 & \\ 0 & 1 & -3 & -5 & \\ 0 & 0 & 0 & 0 & \end{array} \right).$$

Pivots sit in columns 1 and 2; column 3 has none, so z is **free**. Back-substituting, $y = -5 + 3z$ and $x = 4 - 2y - z = 14 - 7z$. The solution set is a line, treated next.

3.3 Pivot analysis and solution classification

After reduction, the pivot pattern decides everything. A column holding a pivot is a **pivot variable**; a column without one is a **free variable**. Then:

- if any row reads $0 = c$ with $c \neq 0$ (a pivot in the augmented column), the system is **inconsistent** — **no solution**;
- otherwise, if **every** variable is a pivot variable, the solution is **unique**;
- otherwise there are free variables, and the system has **infinitely many solutions**, one for each assignment of the free variables.

Common pitfall Count pivots in the coefficient part, but always check the augmented column separately. A pivot in the right-hand column alone signals the contradiction $0 = c$ and means **no solution** — even if the coefficient part looked fine. Many “I found infinitely many solutions” errors are really an overlooked inconsistency.

3.4 Parametric form

When free variables exist, name them as **parameters** and express each pivot variable in terms of them. The solution set then reads as a **particular solution plus a span of directions**:

$$X = X_0 + t_1 V_1 + \dots + t_k V_k, \quad t_1, \dots, t_k \in \mathbb{R},$$

where X_0 is one solution and $V_1, \dots, V_k$ are the directions traced by the free variables.

Worked example — the previous system in parametric form. With $z = t$ free, $x = 14 - 7t$ and $y = -5 + 3t$, so

$$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 14 \\ -5 \\ 0 \end{pmatrix} + t \begin{pmatrix} -7 \\ 3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

This is a line in $\mathbb{R}^3$ through the point $(14, -5, 0)$ in direction $(-7, 3, 1)$. The vector $(-7, 3, 1)$ is exactly the free-variable direction; notice it is what you get by setting $t = 1$ and dropping the constant — this reappears as a **basis of the null space** in Chapter 7.

4 A matrix acting on a vector

We now reinterpret the symbol AX . So far it was a bookkeeping device for a system; now read it as a **machine** that eats a vector and emits a vector. This shift — from “array of numbers” to “transformation” — is the conceptual hinge of the course.

A matrix $A \in M_{m,n}(\mathbb{R})$ defines a map

$$f_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad X \mapsto AX,$$

sending each input column X (with n entries) to the output AX (with m entries). The defining feature of this map is that it **respects linear combinations**:

$$A(X + Y) = AX + AY, \quad A(\lambda X) = \lambda(AX).$$

A map with this property is called **linear** — the central notion of Chapter 10. For now, two subspaces attached to A deserve names, because the entire theory to come is built to compute them.

Definition 4 For $A \in M_{m,n}(\mathbb{R})$:

- the **null space** (or **kernel**) is the set of inputs sent to zero,

$$\ker A = \{X \in \mathbb{R}^n : AX = 0\};$$

- the **column space** (or **image**) is the set of all outputs,

$$\operatorname{im} A = \{AX : X \in \mathbb{R}^n\},$$

which equals the set of all linear combinations of the **columns** of A .

The two are exactly the two sides of solving $AX = B$. The null space is the solution set of the **homogeneous** system $AX = 0$ — the free-variable directions of Chapter 4. The column space is the set of right-hand sides B for which $AX = B$ is **solvable at all**: B is reachable precisely when it is a combination of the columns. Reading “ $AX = B$ has a solution” as “ B lies in the column space” is a habit worth forming now.

Worked example — reading off the column space. For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, both columns point along $(1, 2)$, so every output AX is a multiple of $(1, 2)$: the column space is the **line** spanned by $(1, 2)$, not all of $\mathbb{R}^2$. Hence $AX = B$ is solvable only for B on that line — e.g. solvable for $B = (3, 6)$, impossible for $B = (1, 0)$.

5 Vector spaces

Here is the leap promised in the preface. We have been adding and scaling several different kinds of objects — numerical tuples, polynomials, matrices, functions. We now extract the **rules** they all obey and take those rules as a definition. Anything satisfying them is a **vector space**, and any theorem we prove from the rules alone holds for all of them simultaneously.

5.1 From a familiar example to the axioms

In $\mathbb{R}^n$ you add componentwise and scale componentwise, and these operations satisfy a short list of identities — addition commutes and associates, there is a zero, every vector has a negative, and scalar multiplication distributes. We promote that list to a definition.

Definition 5 A **vector space over $\mathbb{R}$** is a set E equipped with an **addition** $E \times E \rightarrow E$ and a **scalar multiplication** $\mathbb{R} \times E \rightarrow E$ such that, for all $u, v, w \in E$ and all $\lambda, \mu \in \mathbb{R}$:

1. addition is **associative**: $(u + v) + w = u + (v + w)$;
2. addition is **commutative**: $u + v = v + u$;
3. there is a **zero vector** 0 with $u + 0 = u$;
4. every u has an **opposite** $-u$ with $u + (-u) = 0$;
5. $\lambda(u + v) = \lambda u + \lambda v$;
6. $(\lambda + \mu)u = \lambda u + \mu u$;

7. $(\lambda\mu)u = \lambda(\mu u)$;
8. $1 \cdot u = u$.

The elements of E are called **vectors**.

Remark The axioms are a **recognition checklist**, not something to memorize for its own sake. The intended reading is: “ $\mathbb{R}^n$, with the operations you have used since school, satisfies these — so let us agree to call anything else that satisfies them a vector space too, and reuse all our results there.”

5.2 The standard examples

Worked example — the spaces this course lives in. Each of the following is a vector space over $\mathbb{R}$; in each, addition and scaling are the obvious componentwise/pointwise operations, and the zero vector is named.

- $\mathbb{R}^n$ — numerical vectors; zero is $(0, \dots, 0)$.
- $\mathbb{R}[X]$ — all polynomials, and $\mathbb{R}_n[X]$ — polynomials of degree $\leq n$; zero is the zero polynomial.
- $\mathcal{F}(\mathbb{R}, \mathbb{R})$ — all functions $\mathbb{R} \rightarrow \mathbb{R}$, with $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$; zero is the constant function 0.
- $M_{m,n}(\mathbb{R})$ — matrices of a fixed size; zero is the zero matrix.

Common pitfall Watch the degree example. The polynomials of degree **exactly** n do **not** form a vector space: their sum can have lower degree (e.g. $(X^2 + X) + (-X^2) = X$), and they contain no zero polynomial. It is $\mathbb{R}_n[X]$, degree **at most** n , that is a space. “At most” is closed under addition; “exactly” is not.

6 Subspaces

In practice one rarely checks all eight axioms from scratch. Instead, most spaces of interest sit **inside** a space already known to be a vector space — the solutions of $AX = 0$ inside $\mathbb{R}^n$, the polynomials of degree ≤ 2 inside $\mathbb{R}[X]$. For such **subsets**, a three-line test replaces the whole axiom list.

6.1 The three-criterion test

Definition 6 A subset F of a vector space E is a **subspace** of E if it is itself a vector space under the operations of E .

Theorem 7 (Three-criterion test) A subset $F \subseteq E$ is a subspace if and only if

1. $0 \in F$ (it contains the zero vector);
2. F is **closed under addition**: $u, v \in F \implies u + v \in F$;
3. F is **closed under scalar multiplication**: $\lambda \in \mathbb{R}$ and $u \in F \implies \lambda u \in F$.

The test works because the eight axioms are **inherited** automatically from E : associativity, commutativity and the rest already hold for all of E , hence for its elements in F . The only

things that could go wrong are that F misses the zero vector or “leaks” outside itself under the operations — and that is exactly what the three criteria forbid. Criteria 2 and 3 are often merged: F is a subspace iff $0 \in F$ and $\lambda u + v \in F$ for all $\lambda \in \mathbb{R}$, $u, v \in F$.

The geometric content of criterion 1 is worth a picture: a subspace must pass through the origin.

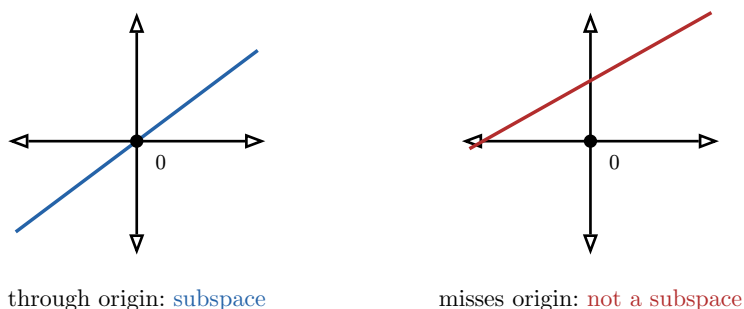


Figure 2: Every subspace contains the zero vector, so in $\mathbb{R}^2$ the only subspaces are $\{0\}$, lines **through the origin**, and the whole plane. A line that misses the origin (right) fails criterion 1 immediately — a quick first test before checking closure.

Worked example — applying the test. Is $F = \{(x, y, z) \in \mathbb{R}^3 : x + y - 2z = 0\}$ a subspace of $\mathbb{R}^3$? (1) $(0, 0, 0)$ satisfies the equation, so $0 \in F$. (2) If (x, y, z) and (x', y', z') both satisfy it, adding the two equations shows their sum does too. (3) Scaling an equation by λ preserves it. All three hold, so F is a subspace. Contrast $G = \{(x, y, z) : x + y - 2z = 1\}$: here $0 \notin G$, so G fails at criterion 1 and is **not** a subspace.

Common pitfall A solution set of a **homogeneous** system $AX = 0$ is always a subspace; a solution set of an **inhomogeneous** system $AX = B$ with $B \neq 0$ never is (it misses the origin). This is exactly the distinction between the null space and a parametric solution $X_0 + (\dots)$.

6.2 Span: the subspace generated by a set

Given some vectors, the smallest subspace containing them is obtained by closing them under the two operations — that is, by forming all their linear combinations.

Definition 8 The **span** (or **generated subspace**) of vectors $v_1, \dots, v_k \in E$ is

$$\text{span}(v_1, \dots, v_k) = \{\lambda_1 v_1 + \dots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

It is a subspace of E , and it is the **smallest** subspace containing all of $v_1, \dots, v_k$.

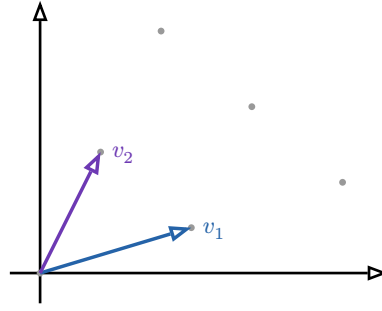


Figure 3: The span of two non-parallel vectors v_1, v_2 in $\mathbb{R}^2$. The grey dots are the combinations $iv_1 + jv_2$ for integer i, j ; allowing **real** coefficients fills in everything between them, so the span is the entire plane. Two non-parallel vectors already generate $\mathbb{R}^2$.

6.3 Column space and null space by row reduction

For a matrix A the column space and null space (Chapter 5) are subspaces, and **one row reduction computes both**:

- the **null space** is read from the parametric solution of $AX = 0$ — the free-variable direction vectors span it;
- the **column space** is spanned by the columns of A sitting in **pivot positions** (the pivot columns of the original A).

This is the first sign that row reduction is the universal computational engine: the abstract questions “what is the null space?” and “what is the column space?” are answered by the elimination you already do by hand.

6.4 Sum, intersection, and direct sum

Definition 9 For subspaces F, G of E :

- the **intersection** $F \cap G = \{u : u \in F \text{ and } u \in G\}$ is a subspace;
- the **sum** $F + G = \{u + v : u \in F, v \in G\}$ is a subspace — the smallest one containing both F and G .

The sum is **direct**, written $F \oplus G$, when $F \cap G = \{0\}$. If moreover $F \oplus G = E$, then F and G are **complementary** (or **supplementary**) subspaces of E .

The point of a direct sum is **unique decomposition**: $F \oplus G = E$ means every $u \in E$ writes in **exactly one** way as $u = u_F + u_G$ with $u_F \in F$, $u_G \in G$. Uniqueness is precisely what $F \cap G = \{0\}$ buys — if some vector lay in both, you could shuffle it between the two parts and the decomposition would not be unique.

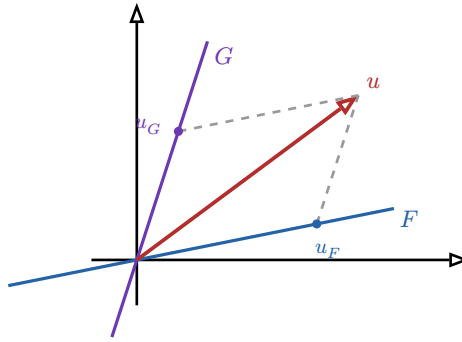


Figure 4: Two distinct lines through the origin are complementary subspaces of $\mathbb{R}^2$: $F \oplus G = \mathbb{R}^2$. Each vector u (red) has a **unique** decomposition $u = u_F + u_G$ found by projecting along the two lines (the dashed parallelogram). Uniqueness fails the instant the lines coincide, i.e. the instant $F \cap G \neq \{0\}$.

7 Linear independence, bases, dimension

We can now answer the question lurking behind “span”: **how many vectors do you really need** to generate a space, and is there redundancy among a given set? The answers — independence, basis, dimension — are the quantitative heart of linear algebra.

7.1 Linear independence

A set of vectors is **redundant** if one of them is a combination of the others; removing it changes nothing about the span. **Independence** is the absence of any such redundancy, captured by a single clean equation.

Definition 10 Vectors $v_1, \dots, v_k$ are **linearly independent** if the only scalars satisfying

$$\lambda_1 v_1 + \dots + \lambda_k v_k = 0$$

are $\lambda_1 = \dots = \lambda_k = 0$ (the **trivial** combination). If some non-trivial combination gives 0, they are **linearly dependent**.

Worked example — testing independence is solving a homogeneous system. Are $v_1 = (1, 1, 0)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 0, -1)$ independent? Set $\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = 0$; componentwise this is the homogeneous system with these vectors as **columns**. Row-reducing finds $\lambda_1 = \lambda_3$, $\lambda_2 = -\lambda_3$, with λ_3 free — a non-trivial solution (take $\lambda_3 = 1$). So they are **dependent**; indeed $v_1 - v_2 - v_3 = 0$. Independence questions are always homogeneous systems in disguise.

Common pitfall Independence is a property of the **whole family**, not of pairs. Three vectors can be pairwise non-parallel yet still dependent — as above, where no two are multiples of each other but the three together satisfy $v_1 - v_2 - v_3 = 0$. Always test the full combination, never just “are any two parallel?”.

7.2 Bases and dimension

Definition 11 Let E be a vector space.

- A family is **generating** (or **spanning**) if its span is all of E .

- A **basis** of E is a family that is **both** linearly independent **and** generating.

A basis is the “just right” family: generating means **enough** vectors to reach everything, independent means **no more than enough**. Equivalently — and this is the property to remember — a basis is a family in which **every** vector of E writes **uniquely** as a linear combination of the basis vectors.

Theorem 12 (Dimension is well-defined) In a finitely generated vector space, all bases have the **same** number of vectors. This common number is the **dimension** of E , written $\dim E$.

Worked example — standard dimensions. The **standard basis** of $\mathbb{R}^n$ is $e_1, \dots, e_n$ with e_i having a 1 in position i and 0 elsewhere; thus $\dim \mathbb{R}^n = n$. Likewise $\dim \mathbb{R}_n[X] = n + 1$ (basis $1, X, \dots, X^n$) and $\dim M_{m,n}(\mathbb{R}) = mn$.

7.3 Reading dimension off the pivots

The connection to Chapter 4 is the practical payoff of the whole course. Form the matrix whose columns are your vectors and row-reduce:

Row reduction reads independence, basis, and dimension.

- the vectors are **independent** iff **every** column is a pivot column (no free variables);
- the **pivot columns of the original matrix** form a basis of their span;
- the **rank** of the matrix — the number of pivots — equals the **dimension** of that span.

So “extract a basis from a generating family” is just “keep the pivot columns”, and “compute the dimension of a span” is just “count the pivots”. The abstract notions of basis and dimension are computed by elimination.

Proposition 13 (Counting in a space of known dimension) In a space E with $\dim E = n$:

- any linearly independent family has **at most** n vectors;
- any generating family has **at least** n vectors;
- any family of **exactly** n vectors that is independent **or** generating is automatically a **basis**.

The last point saves enormous work: once you know $\dim E = n$, a candidate basis of n vectors needs only **one** of the two properties checked, not both.

8 The Four Dimensions Theorem

Sums and intersections of subspaces interact through a single, memorable counting law — the linear-algebra analogue of inclusion–exclusion for sets.

Theorem 14 (Four Dimensions) For finite-dimensional subspaces F, G of a vector space,

$$\dim(F + G) = \dim F + \dim G - \dim(F \cap G).$$

The intuition mirrors counting a union of two sets: adding $\dim F$ and $\dim G$ counts the overlap $F \cap G$ **twice**, so we subtract it once. The theorem is the tool for finding the dimension of an intersection that is awkward to describe directly: rearrange to $\dim(F \cap G) = \dim F + \dim G - \dim(F + G)$.

Worked example — two planes in space. Let F, G be two distinct planes through the origin in $\mathbb{R}^3$, so $\dim F = \dim G = 2$. Their sum is all of $\mathbb{R}^3$ (two distinct planes together reach everything), so $\dim(F + G) = 3$. Then

$$\dim(F \cap G) = 2 + 2 - 3 = 1 :$$

two distinct planes through the origin meet in a **line**, exactly as geometry insists.

Proposition 15 (Direct sum, by the numbers) $F \oplus G = E$ holds if and only if both $F \cap G = \{0\}$ and $\dim F + \dim G = \dim E$.

This is the practical test for complementary subspaces: check that the dimensions add up to $\dim E$ and that the intersection is trivial. By the Four Dimensions Theorem the two conditions together force $\dim(F + G) = \dim E$, so the sum fills E .

Common pitfall Both conditions are needed. Two distinct lines in $\mathbb{R}^3$ satisfy $F \cap G = \{0\}$ but only $\dim F + \dim G = 2 \neq 3$, so they do **not** span $\mathbb{R}^3$. Conversely the dimensions can add up while the intersection is non-trivial. Neither condition alone certifies a direct-sum decomposition.

9 Linear maps

We promoted the rules of $\mathbb{R}^n$ to the abstract notion of a vector space. Now we do the same for the **maps** between them. The map $X \mapsto AX$ of Chapter 5 respected linear combinations; we take that property as the definition of the maps worth studying.

9.1 Definition, kernel, image

Definition 16 A map $f : E \rightarrow F$ between vector spaces is **linear** if for all $u, v \in E$ and $\lambda \in \mathbb{R}$,

$$f(u + v) = f(u) + f(v), \quad f(\lambda u) = \lambda f(u).$$

Equivalently, f commutes with all linear combinations: $f(\lambda u + \mu v) = \lambda f(u) + \mu f(v)$.

Worked example — linear and non-linear. The map $f(x, y) = (2x - y, x + 3y)$ on $\mathbb{R}^2$ is linear — each output coordinate is a linear combination of the inputs, so $f(X) = AX$ with $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$. The map $g(x, y) = (x + 1, y)$ is **not** linear: it sends 0 to $(1, 0) \neq 0$, and a linear map must fix the zero vector (take $\lambda = 0$). Differentiation $p \mapsto p'$ on $\mathbb{R}_n[X]$ is linear; so is integration $f \mapsto \int_0^1 f$. Linearity is everywhere once you look.

Common pitfall A linear map **must** send 0 to 0: setting $\lambda = 0$ gives $f(0) = 0$. So any map with a non-zero constant term — an “affine” map like $x \mapsto ax + b$ with $b \neq 0$ — is **not**

linear, despite the school usage of “linear” for straight lines. Checking $f(0) = 0$ is the fastest disqualifier.

Definition 17 For a linear map $f : E \rightarrow F$:

- the **kernel** is $\ker f = \{u \in E : f(u) = 0\} \subseteq E$;
- the **image** is $\operatorname{im} f = \{f(u) : u \in E\} \subseteq F$;
- the **rank** is $\operatorname{rk} f = \dim(\operatorname{im} f)$.

Both $\ker f$ and $\operatorname{im} f$ are subspaces (of E and F respectively).

These generalize the null space and column space of a matrix verbatim: for $f(X) = AX$, $\ker f = \ker A$ and $\operatorname{im} f = \operatorname{im} A$. Everything you learned to compute by row reduction transfers directly.

9.2 The rank theorem

The single most important theorem about a linear map says its domain splits cleanly into “what gets crushed to zero” and “what survives as the image”.

Theorem 18 (Rank theorem (rank-nullity)) For a linear map $f : E \rightarrow F$ with E finite-dimensional,

$$\dim E = \dim(\ker f) + \operatorname{rk} f.$$

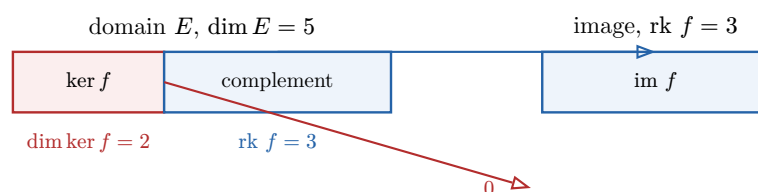


Figure 5: The rank theorem as a budget. The domain E (here $\dim E = 5$) splits into the part crushed to zero — $\ker f$, of dimension 2 — and a complement that maps **bijectively** onto the image, of dimension $\operatorname{rk} f = 3$. The dimensions must add up: $5 = 2 + 3$. Enlarging the kernel necessarily shrinks the rank, and vice versa.

The theorem is a conservation law: the $\dim E$ “degrees of freedom” in the domain are spent either on directions that vanish (the kernel) or on directions that survive into the image (the rank), with none lost. For a matrix $A \in M_{m,n}(\mathbb{R})$ it reads $n = \dim(\ker A) + \operatorname{rk} A$, which is just the pivot/free-variable split of Chapter 4: rank = number of pivots, nullity = number of free variables, and together they account for all n columns.

9.3 Injectivity, surjectivity, bijectivity

Linearity turns the function-theoretic notions of MAT11-1 into **dimension counts** — this is where the abstraction pays its way.

Proposition 19 Let $f : E \rightarrow F$ be linear with E, F finite-dimensional.

- f is **injective** $\iff \ker f = \{0\} \iff \operatorname{rk} f = \dim E$.
- f is **surjective** $\iff \operatorname{im} f = F \iff \operatorname{rk} f = \dim F$.
- f is **bijective** $\iff$ both hold; this forces $\dim E = \dim F$.

The injectivity criterion is the workhorse: instead of checking “ $f(u) = f(v) \implies u = v$ ” for all pairs, you check the **single** equation $f(u) = 0 \implies u = 0$. The reason is linearity — $f(u) = f(v)$ rearranges to $f(u - v) = 0$, so distinct inputs collide exactly when the kernel is non-trivial.

Proof (injective $\iff$ trivial kernel). If f is injective then $f(u) = 0 = f(0)$ forces $u = 0$, so $\ker f = \{0\}$. Conversely, suppose $\ker f = \{0\}$ and $f(u) = f(v)$. By linearity $f(u - v) = f(u) - f(v) = 0$, so $u - v \in \ker f = \{0\}$, giving $u = v$. Hence f is injective. $\square$

Corollary 20 For a linear map $f : E \rightarrow F$ between spaces of **equal finite dimension** ($\dim E = \dim F$), injectivity, surjectivity and bijectivity are **equivalent**.

Proof. By the rank theorem $\dim(\ker f) = \dim E - \operatorname{rk} f$. Injectivity means $\dim(\ker f) = 0$, i.e. $\operatorname{rk} f = \dim E = \dim F$, which is exactly surjectivity. So each $\implies$ the other, and either $\implies$ bijectivity. $\square$

Common pitfall That equivalence holds **only when the dimensions are equal** (in particular for a square matrix). In general a tall thin map can be injective without being surjective, and a short fat one surjective without being injective. Always confirm $\dim E = \dim F$ before invoking “injective $\implies$ bijective”.

10 Matrix representation and change of basis

We close the circle. We began with matrices, abstracted to linear maps, and now show that **every** linear map between finite-dimensional spaces is, once bases are chosen, **exactly a matrix**. The abstract and the concrete are two views of one thing.

10.1 Coordinate vectors

Definition 21 Let $\mathcal{B} = (e_1, \dots, e_n)$ be a basis of E . Each $u \in E$ writes uniquely as $u = x_1 e_1 + \dots + x_n e_n$; the column

$$[u]_{\mathcal{B}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is the **coordinate vector** of u in the basis $\mathcal{B}$.

The uniqueness in the definition is precisely the basis property; it is what makes coordinates well-defined. Choosing a basis is choosing a **dictionary** that translates every abstract vector — a polynomial, a function — into a concrete column of numbers.

Worked example — coordinates depend on the basis. In $\mathbb{R}_2[X]$ with basis $\mathcal{B} = (1, X, X^2)$, the polynomial $p = 3 - X + 2X^2$ has coordinate vector $[p]_{\mathcal{B}} = (3, -1, 2)^T$. In a **different** basis the same p gets a **different** column — coordinates are a property of the (vector, basis) pair, never of the vector alone.

10.2 The matrix of a linear map

Definition 22 Let $f : E \rightarrow F$ be linear, $\mathcal{B} = (e_1, \dots, e_n)$ a basis of E and $\mathcal{C}$ a basis of F . The **matrix of f relative to $\mathcal{B}$ and $\mathcal{C}$** , written $\text{Mat}_{\mathcal{B}, \mathcal{C}}(f)$, is the matrix whose **j -th column** is the coordinate vector $[f(e_j)]_{\mathcal{C}}$ of the image of the j -th basis vector.

The rule “columns are the images of the basis vectors” is worth committing to memory; it is **the** recipe for building the matrix. Its whole purpose is the following compatibility: computing with the abstract map and computing with its matrix give the same answer.

Proposition 23 With $A = \text{Mat}_{\mathcal{B}, \mathcal{C}}(f)$: if $[u]_{\mathcal{B}} = X$ then $[f(u)]_{\mathcal{C}} = AX$. Applying f to a vector becomes multiplying its coordinate column by A .

Worked example — matrix of differentiation. Let $f = \frac{d}{dX}$ on $\mathbb{R}_2[X]$, basis $\mathcal{B} = (1, X, X^2)$ for both source and target. The images of the basis vectors are $f(1) = 0$, $f(X) = 1$, $f(X^2) = 2X$, with coordinate columns $(0, 0, 0)^T$, $(1, 0, 0)^T$, $(0, 2, 0)^T$. Stacking them as columns,

$$\text{Mat}_{\mathcal{B}}(f) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check: $p = 3 - X + 2X^2$ has $[p] = (3, -1, 2)^T$, and $A(3, -1, 2)^T = (-1, 4, 0)^T$, the coordinates of $-1 + 4X = p'$. ✓

10.3 Change of basis

A vector does not change when we re-describe it in a new basis, but its coordinate column does. The bookkeeping is handled by a single matrix.

Definition 24 Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of E . The **change-of-basis matrix P** from $\mathcal{B}$ to $\mathcal{B}'$ has, as its columns, the coordinate vectors of the **new** basis vectors $\mathcal{B}'$ expressed in the **old** basis $\mathcal{B}$.

Proposition 25 If $X = [u]_{\mathcal{B}}$ and $X' = [u]_{\mathcal{B}'}$ are the coordinates of the same vector u in the two bases, then

$$X = PX'.$$

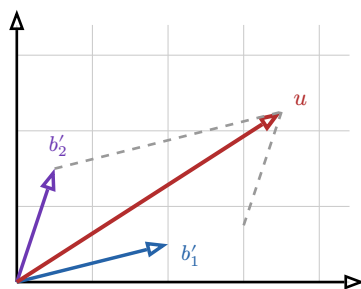


Figure 6: One vector u (red), two readings. Against the standard grid its coordinates are read off the horizontal/vertical axes; against the skewed basis (b'_1, b'_2) (blue, purple) the **same** u has coordinates from the dashed parallelogram — here $u = 1.5b'_1 + 1b'_2$. The change-of-basis matrix

P converts one column of numbers into the other; the arrow itself never moves.

Change of basis is introduced here and developed further in **Linear Algebra II**, where it explains how the matrix of a single map transforms when we change the bases — the gateway to diagonalization.

The arc in one breath. A system $AX = B$ is solved by **row reduction**; its pivots and free variables become **rank** and **nullity**. The same matrix is a **linear map**, whose kernel and image are the null space and column space. Abstracting the rules of $\mathbb{R}^n$ gives **vector spaces**; the smallest honest generating-and-independent family is a **basis**, its size the **dimension**, both **read off the pivots**. The rank theorem $\dim E = \dim \ker f + \text{rk } f$ ties domain, kernel and image together, and choosing bases turns every linear map back into a matrix — closing the circle we opened.